

Benchmarking Language Models for Statistical Problem Formulation

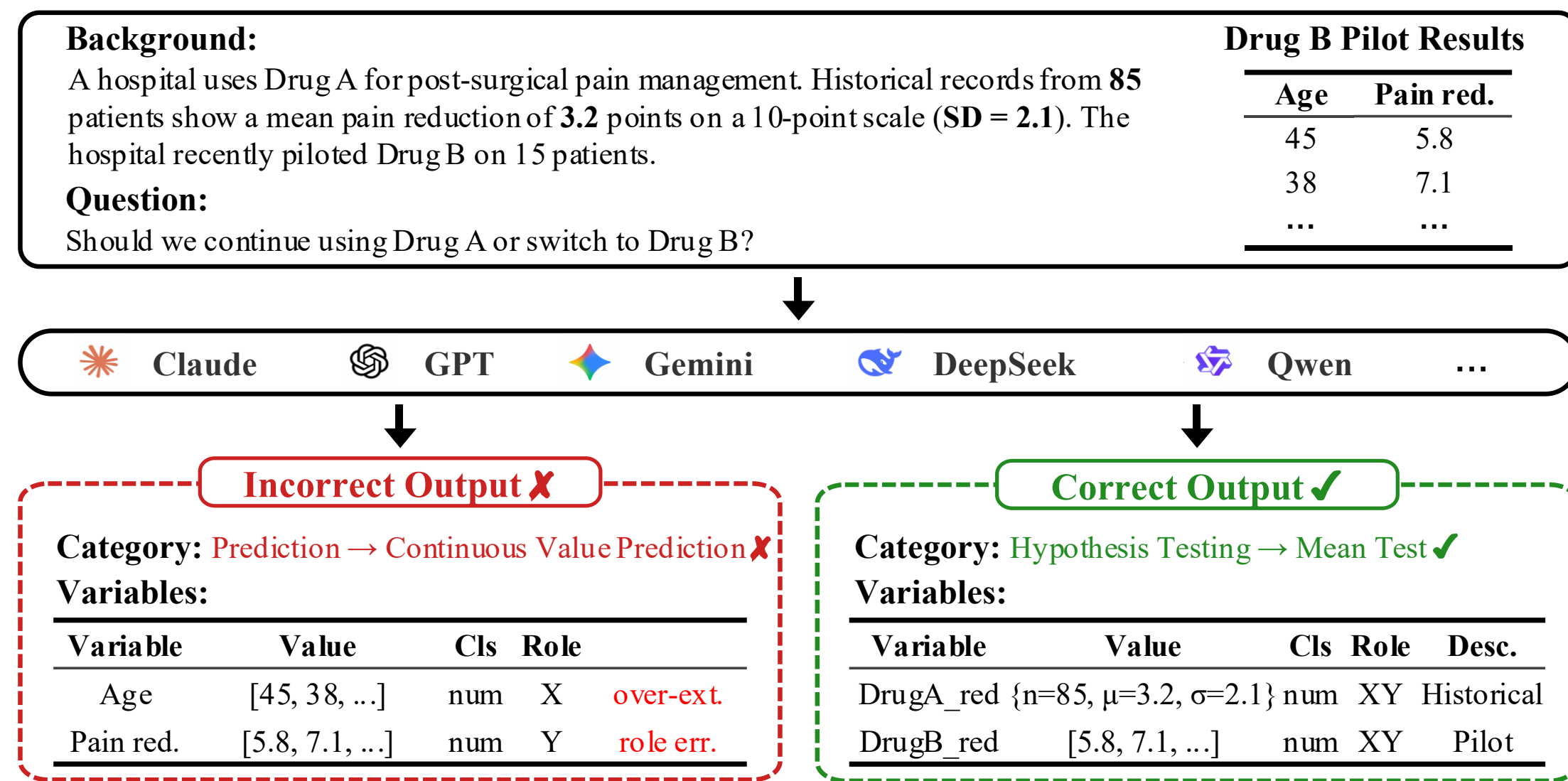
Chen Wang^{1*}, Junzhe Zhao^{1*}, Xin Cong^{1†}, Wanlu Deng¹, Ke Deng¹

¹ Department of Statistics and Data Science, Tsinghua University



Motivation and Background

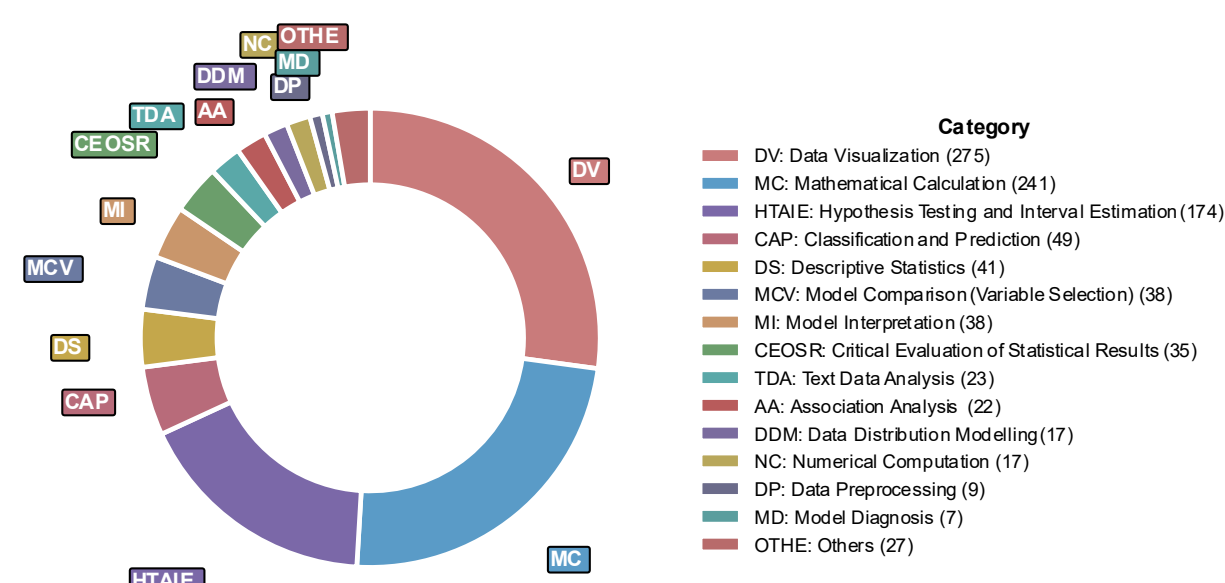
- Large language models (LLMs) are increasingly used as assistants for statistical and data science work, yet existing evaluations largely assume the **analysis target is already specified**.
- In practice, users arrive with informal goals and heterogeneous data, leaving the model to decide what statistical task is implied and which data are relevant. These implicit choices determine whether the subsequent analysis is statistically valid. We refer to this upstream specification step as **statistical problem formulation**.
- Existing benchmarks assume that the analysis target is **already given or strongly implied**, so they do not directly test whether a model can **formulate the statistical task** from an informal real-world request.



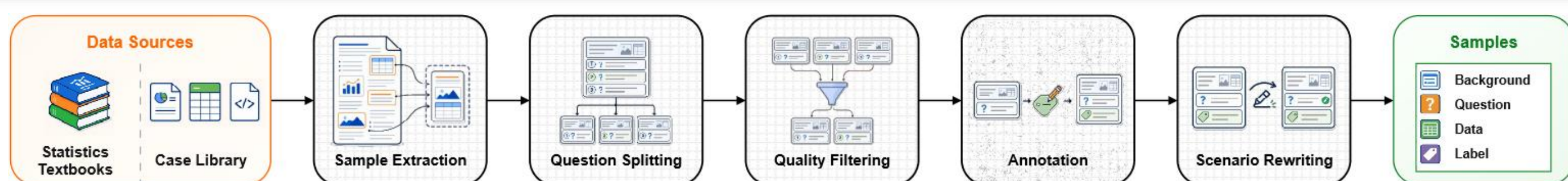
Task Formulation

We define Statistical Problem Formulation as two subtasks:

- Statistical Problem Classification:**
 $(B, Q, D) \rightarrow c = (c_{CG}, c_{FG})$
- Variable Identification and Role Assignment:**
 $(B, Q, D) \rightarrow v_i = (id_i, desc_i, value_i, role_i)$
- Final taxonomy:** 20 **coarse-grained** and 85 **fine-grained** statistical problem categories.
- Roles:** one of four values: **predictor, response, both, or N/A**.



Benchmark Construction: StatFormBench



Data sources: five statistics textbooks and a case-based data science library from GouXiongHui.

- Sample extraction and splitting:** Raw materials are first parsed into entries, and entries with multiple sub-questions are then split into individual samples by extracting the shared background and pairing each sub-question with its answer.
- Quality filtering:** Samples that are too narrow to test formulation, are removed, and cases with incomplete information are discarded.
- Initial label construction:** Each retained sample is annotated with a category and a variable set, with model-generated labels serving only as initial candidates.
- Scenario rewriting:** GPT-5 rewrites each problem from the perspective of a domain practitioner.
- Human verification:** All 1,013 samples are manually reviewed, with the problem category c and the variable set V both verified against the reference answers in the source materials.

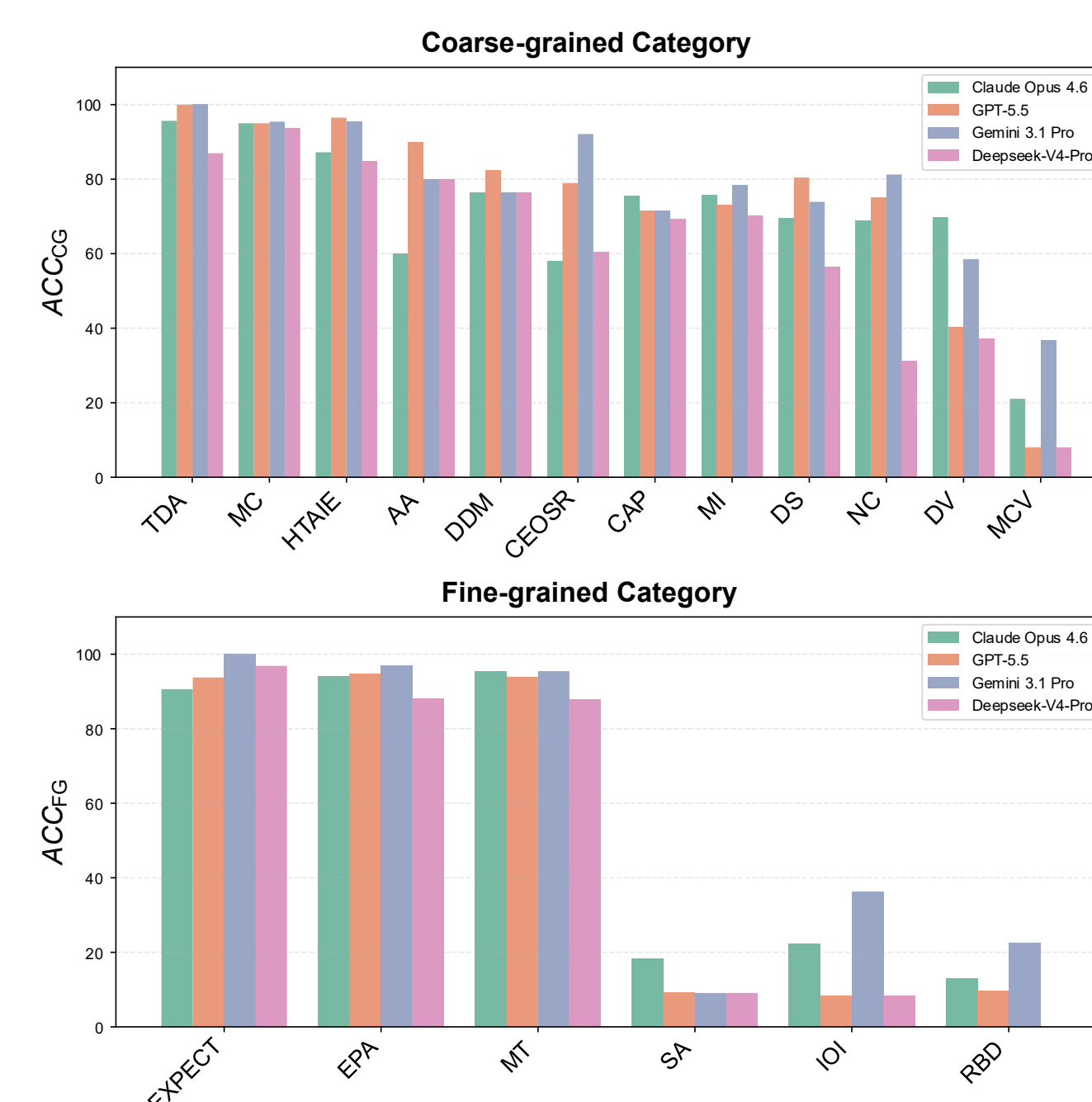
Experimental Results

Metrics:

Metric	Description
ACC _{CG}	Coarse-grained accuracy
ACC _{FG}	Fine-grained accuracy
JCV	Jaccard Coefficient measuring variable set similarity
PV	Precision: proportion of correct predicted variables
RV	Recall: proportion of correct variables that are predicted
VRI	Variable Role Consistency: matching variable roles

Model	Classification		Variable Id. & Role			
	ACC _{CG}	ACC _{FG}	JCV	PV	RV	VRI
Claude Opus 4.6	77.7	69.5	63.2	65.8	72.9	66.2
Claude Sonnet 4.6	71.9	63.9	62.4	65.0	70.9	63.3
GPT-5.5	72.9	67.2	52.1	53.7	67.2	61.1
GPT-5.4	75.2	65.8	52.0	53.7	68.3	62.0
Gemini 3.1 Pro	79.0	72.0	61.5	64.5	68.3	63.1
Gemini 3 Flash	77.0	68.4	62.4	66.1	68.9	63.6
Kimi-K2.5	71.5	64.8	59.3	61.3	69.8	63.8
DeepSeek-V4-Pro	66.2	59.1	61.5	64.9	67.3	60.3
DeepSeek-V4-Flash	71.2	64.8	54.5	56.1	70.9	65.3
Qwen3.5-397B-A17B	70.6	64.6	57.2	60.5	66.5	61.4
Qwen3.5-9B	29.1	20.3	48.3	51.6	58.1	45.0
Qwen3.5-4B	37.7	30.6	45.4	48.7	54.9	41.1
Qwen3.5-2B	7.7	3.5	22.3	32.0	25.2	7.1
Qwen3.5-0.8B	7.9	3.8	5.6	8.3	7.6	3.0

- Main results:** Across 14 open- and closed-source LLMs, the best zero-shot models reach only **72.0** fine-grained classification accuracy and **63.2** variable set overlap.
- Per-category performance:** Models can capture the broad analytical intent but still struggle to distinguish closely related subtypes.
- Data source impact:** On textbook samples, fine-grained accuracy ranges from **74.1** to **84.1** while JCV ranges from **33.6** to **51.0**; on case-library samples, **the pattern reverses**.
- Prompting strategies:** Providing category definitions improves fine-grained classification for all three models by 1.9 to 3.8 points, but effects on variable identification and role assignment are smaller and mixed. Overall, few-shot examples **do not substantially improve** statistical problem formulation.



Source	Model	ACC _{FG}	JCV	VRI
Book	Claude Opus 4.6	78.5	51.0	56.9
	Gemini 3.1 Pro	83.8	48.1	50.9
	GPT-5.5	84.1	33.6	47.5
	DeepSeek-V4-Pro	74.1	48.6	50.0
Case	Claude Opus 4.6	54.7	83.2	81.3
	Gemini 3.1 Pro	52.6	83.6	83.1
	GPT-5.5	39.6	82.4	83.3
	DeepSeek-V4-Pro	34.6	82.5	77.2

Model	ACC _{FG}	JCV	VRI
Claude Opus 4.6 w/ Category Def.	69.5	63.2	66.2
GPT-5.5 w/ Category Def.	67.2	52.1	61.1
Gemini 3.1 Pro w/ Category Def.	72.0	61.5	63.1

Conclusion

- Experiments on 14 LLMs show that **no model performs consistently best** across the two formulation tasks.
- These findings indicate that statistical problem formulation remains beyond the reliable reach of current models. Bridging this gap may require **tighter integration of domain knowledge, statistical reasoning, and data-aware inference**.
- We hope StatFormBench helps clarify where current models fall short and supports the development of more capable statistical formulation assistants.



wangchen_23@mails.tsinghua.edu.cn
zhaojz26@mails.tsinghua.edu.cn
congxin1995@tsinghua.edu.cn



<https://github.com/THU-CongLab/StatFormBench>



<https://huggingface.co/datasets/THU-CongLab/StatFormBench>